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Project Report

CHRONO_01 – Giant 7-Segment Digital Chronometer

Comprehensive Electrical Study, High-Voltage 24V PCB Design, NPTH Rapid-Prototyping Optimization, and Dual IoT/GSM Telemetry

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ATTESTATION DE STAGE

Je soussigné, Mr. **Mohamed Amine BEN AYED**, agissant en tant que président-directeur général de la société Chrono Sports Events, confirme que
Mr **Bacem Ameur** a effectué un stage au sein de notre entreprise du 19/06/2026 au 19/07/2026.

Cette attestation est délivrée à l'intéressé pour servir et valoir ce que de droit.

Fait à Sfax, le 30 septembre 2026.

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Abstract

Keywords: Embedded Systems, Custom PCB Engineering, KiCad 10, ESP32-WROVER-E, 74HC595 Shift Registers, ULN2803A Darlington Arrays, SIM800L Cellular Engine, 24V High-Power LED Displays, Non-Plated Through Hole (NPTH), Power Transient Sizing, Asynchronous Web Dashboard, Optical Diffuser.

Context: Large-scale outdoor athletic events, automotive rallies, and industrial timing installations require large-format numerical displays capable of delivering direct sunlight visibility, high operational robustness, and flexible multi-protocol control interfaces. Commercial turnkey timing boards are typically high in cost and utilize closed proprietary wireless systems that lack remote worldwide connectivity.

Problem Statement: Developing a custom, cost-effective giant chronometer system involves integrating high-voltage, high-current 24 V display hardware with sensitive 3.3 V/5 V low-power IoT microcontrollers. This creates complex multidisciplinary engineering trade-offs: sinking up to 6.3 A of aggregate display current without thermal degradation, buffering 2.0 A transient RF bursts from GSM transceivers to prevent microcontroller brownout resets, and adapting double-sided PCB layout topologies to accommodate severe rapid-prototyping constraints (specifically Non-Plated Through Holes, track clearances ≥ 0.38 mm, and manual soldering access).

Methodology: This project report documents the comprehensive design, mathematical component sizing, CAD layout optimization, firmware development, physical chassis construction, and experimental validation of **CHRONO_01**. The system architecture utilizes an ESP32-WROVER-E SoC hosting an asynchronous local HTTP Web server alongside a SIM800L GSM engine. A serial-to-parallel SPI cascade composed of six daisy-chained 74HC595 shift registers interfaces six ULN2803A Darlington transistor arrays to sink current from 42 individual 24 V LED strip segments. The PCB layout underwent a rigorous spatial redesign under **KiCad 10**, reducing board area by 58.4% (from 280×110 mm to 160×80 mm) through a novel modular orthogonal "T-Shape" digit block geometry, teardrop pad fillets, and strategic through-hole elevation using machined-pin tulip sockets.

Results: The optimized PCB design achieved 100% DRC validation with zero electrical violations. Physical prototyping and oscilloscope testing confirmed transient voltage ripple suppression below 80 mV during GSM transmission following the integration of an engineered 1000 μF buffer capacitor. The system was housed within a custom matte black enclosure equipped with a 3 mm Opal White PMMA diffuser, achieving sharp, continuous digit illumination without optical bleed.

Conclusion: The final platform delivers an industrial-grade, field-deployable chronometer with hybrid local WiFi and worldwide cellular SMS control. The report concludes with detailed roadmaps for future enhancements, including pin-compatible 4G LTE-Cat1 migration and wide-input automotive power regulation.

Dedication

To my beloved parents and family, for their endless sacrifices, unwavering encouragement, and profound faith in my aspirations.

To the faculty and engineering professors of ENIM, who cultivate technical rigor, innovation, and scientific integrity.

To all engineering students and makers who transform theoretical equations into living, tangible technology.

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List of Acronyms

Acronym	Definition
AC	Alternating Current
ADC	Analog-to-Digital Converter
API	Application Programming Interface
BJT	Bipolar Junction Transistor
BOM	Bill of Materials
CMOS	Complementary Metal-Oxide-Semiconductor
DC	Direct Current
DIP	Dual In-line Package
DRC	Design Rule Check
DRL	Drill File (NC Excellon format)
GPIO	General Purpose Input/Output
GSM	Global System for Mobile Communications
HTTP	Hypertext Transfer Protocol
IC	Integrated Circuit
IoT	Internet of Things
LTE	Long-Term Evolution
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
NPTH	Non-Plated Through Hole
NTP	Network Time Protocol
PCB	Printed Circuit Board
PTH	Plated Through Hole
PWM	Pulse-Width Modulation
REST	Representational State Transfer
RTC	Real-Time Clock
SMD	Surface-Mount Device
SPI	Serial Peripheral Interface
VCC	Voltage Common Collector

Chapter 1

Project Context, Industrial Requirements and System Architecture

1.1 Introduction

Large-format electronic timing displays and scoreboards are vital across professional athletics, vehicular rallies, public events, and industrial monitoring. The principal function of such displays is to convey high-precision chronological data across substantial distances under demanding optical and environmental conditions.

This project, titled **CHRONO_01**, details the complete engineering workflow involved in designing, simulating, laying out, fabricating, programming, and validating an industrial-grade 6-digit giant 7-segment chronometer ($HH : MM : SS$). The system is driven by high-intensity 24 V LED strip arrays and incorporates dual wireless control modalities: a local WiFi-hosted Web dashboard and worldwide SMS cellular telemetry.

1.2 Context and Motivation

Outdoor athletic competitions—such as circuit racing, marathons, and collegiate sporting events—present stringent requirements for timing instrumentation:

1. **Long-Distance Optical Legibility:** Displays must maintain crisp contrast and high luminous output under direct solar illumination ($> 50\,000$ lux ambient daylight), requiring high-power 24 V LED bars rather than low-power discrete 5V LEDs.

2. **Deployment Flexibility:** Operators on the track or sports field need instantaneous control via smartphone or tablet without installing proprietary native apps or relying on external internet routers.

3. **Remote Telemetry:** For point-to-point off-road rallies or remote roadside installations, the system must execute start, stop, and split-time commands across arbitrarily

long distances via cellular infrastructure.

4. **Production Cost and Modularity:** Turnkey commercial displays often retail for thousands of dinars and utilize proprietary controller architectures. Developing an open, repairable, highly optimized custom PCB provides substantial economic and educational value.

1.3 Technical and Functional Specifications

To establish an engineering benchmark, the requirements were formalized in Table 1.1.

Table 1.1: CHRONO_01 Technical and Functional Specifications

Parameter	**Specification / Requirement**
Display Format	6 Numerical Digits ($HH:MM:SS$) with dual colon separators (:)
Total Segment Count	42 Active LED Bar Segments (6 digits $\times$ 7 segments)
Operating Voltage	24 V DC (Main Power Bus) / 5 V DC (Logic & GSM) / 3.3 V (MCU)
Digit Height / Visibility	$\approx 150 - 200$ mm per digit / Readable from > 50 meters
Maximum Power Draw	≈ 45 W (24 V@1.875 A) with all segments illuminated (`88:88:88`)
Local Control Interface	Embedded Asynchronous HTTP Web Server (802.11b/g/n Soft-AP)
Remote Control Interface	Cellular SMS Telemetry via AT Command Parsing Engine
PCB Form Factor	Compact dual-layer board ($\leq 160 \times 80$ mm), single-chassis mounting
Fabrication Standard	NPTH-compatible rapid prototyping, trace width $\geq$ 0.38 mm

1.4 System Architecture and Data Flow

The architecture is structured into four distinct physical and electrical domains:

- **Power Conditioning Domain:** Accepts 24 V DC from an external AC/DC SMPS or automotive boost converter, distributing 24 V directly to display anodes while an on-board switching Buck converter generates a regulated 5 V rail.

- **Central Processing Domain:** The ESP32 dual-core SoC coordinates timing, parses incoming cellular messages, hosts the web dashboard, and shifts binary display frames.
- **Logic and Driver Domain:** Six cascaded 74HC595 shift registers serialize 48 bits of segment data, which are level-shifted and current-sunk by six ULN2803A Darlington driver ICs.
- **Display Enclosure Domain:** 42 precision-cut 24V LED strip bars housed behind optical light baffles and an opal white acrylic diffusion panel.

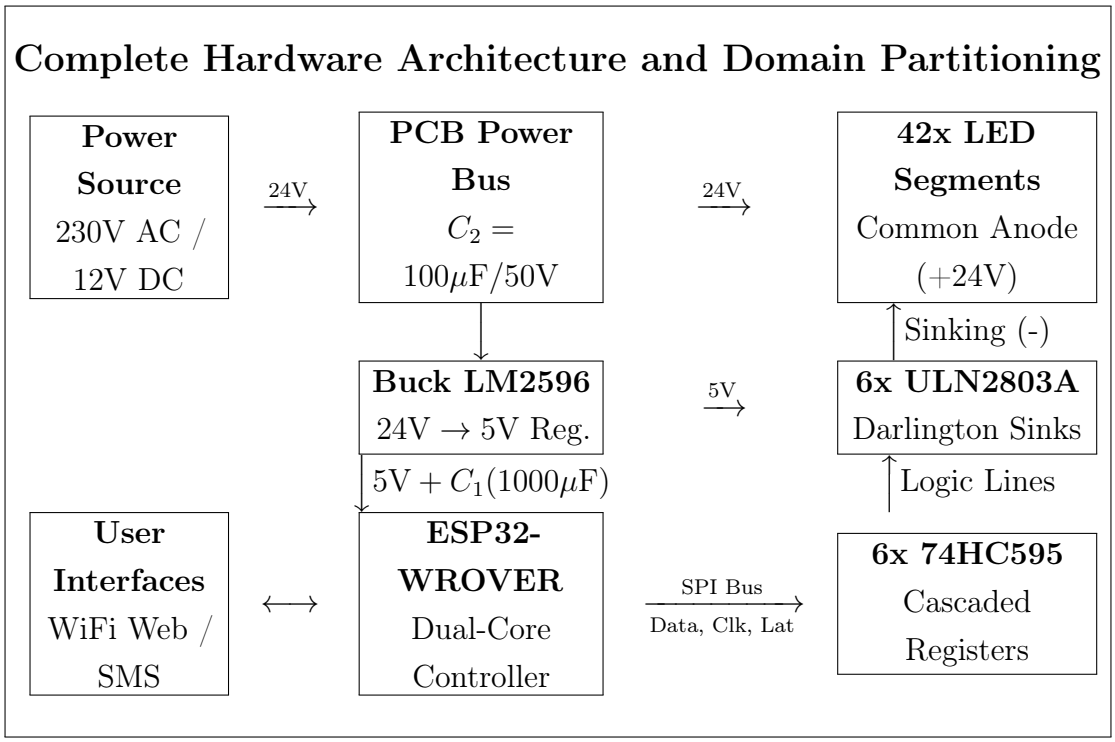


Figure 1.1: CHRONO_01 Detailed System Architecture and Inter-Domain Signal Paths

Chapter 2

Preliminary Comparative Study and Hardware Sizing

2.1 Introduction

Designing an industrial embedded system requires evaluating alternative components to identify optimal trade-offs between performance, reliability, ease of integration, and procurement feasibility. This chapter presents the comparative study that justified our component selections, followed by the rigorous mathematical sizing of the electrical network.

2.2 Component Selection and Trade-Off Analysis

2.2.1 Microcontroller Unit (MCU) Benchmarking

Three candidate processing platforms were evaluated and compared against the chosen Espressif **ESP32-WROVER-E**: the 8-bit ATmega328P (widely used in Arduino Uno boards) and the 32-bit ARM Cortex-M4 STM32F401.

Table 2.1: Microcontroller Benchmarking Matrix

Feature / Parameter	**ATmega328P**	**STM32F401**	**ESP32-WROVER-E** (*Selected*)
Architecture	8-bit AVR	32-bit ARM M4	**32-bit Dual-Core Xtensa**
Clock Frequency	16 MHz	84 MHz	240 **MHz**
SRAM	2 KB	96 KB	520 **KB** + 8 **MB PSRAM**
Flash Memory	32 KB	512 KB	4 **MB SPI Flash**
Integrated Wi-Fi	None	None	802.11**b/g/n Native**
Integrated Bluetooth	None	None	**BLE 4.2 Native**
Hardware UART Channels	1	3	**3 Hardware UARTs**
Asynchronous Web Server	Unfeasible	Complex (Shield)	**Native AsyncTCP HTTP**
Dual-Core Parallelism	No	No	**Yes (Core0 + Core1)**
Unit Cost (Tunisia, 2025/2026)	≈ 15 TND	≈ 35 TND	≈ 32 **TND**

Decision Rationale: The ESP32-WROVER-E was selected as the only platform capable of natively hosting both a dual-core parallel real-time execution environment and integrated WiFi without any external shields. Its dual-core architecture allows Core 0 to handle the network stack (Soft-AP and asynchronous HTTP Web server) while Core 1 executes deterministic 1 ms chronometer tick counting, SPI display frame updates, and UART GSM AT command parsing without timing jitter. The ATmega328P was disqualified by its lack of WiFi and insufficient SRAM for HTML page hosting. The STM32F401, while computationally capable, would require an external WiFi module (ESP8266 or W5500 Ethernet shield), adding cost and PCB footprint.

2.3 Economic Cost Analysis: Custom Development vs. Commercial Purchase

2.3.1 Overview

A critical engineering decision at the project outset was determining whether to design and build a custom chronometer or acquire an equivalent commercial product. This section presents the detailed financial comparison between the cost of the custom **CHRONO_01** platform and the market price of equivalent turnkey commercial timing systems.

2.3.2 CHRONO_01 Bill of Materials Cost (Detailed Breakdown)

All components were sourced from Tunisian suppliers (Didactico Tunis, CoThings, GeekShop.tn, local Sfax electronic shops) with 2025/2026 pricing.

Table 2.2: Detailed Component Cost Breakdown for CHRONO_01 (Tunisian Market)

Component / Item	**Qty**	**Unit Price (TND)**	**Subtotal (TND)**
Electronic Components			
ESP32-WROVER-E Development Board	1	32.00	32.00
74HC595 Shift Register IC (DIP-16)	6	2.50	15.00
Machined Tulip DIP-16 Sockets (paire)	6	1.80	10.80
ULN2803A Darlington Array IC (DIP-18)	6	3.20	19.20
Machined Tulip DIP-18 Sockets (paire)	6	2.00	12.00
SIM800L GSM Module + SIM Card	1	45.00	45.00
LM2596 Buck Step-Down Module (24V $\rightarrow$ 5V)	1	12.00	12.00
Electrolytic Capacitor 1000 μF/16V (low-ESR)	2	2.50	5.00
Electrolytic Capacitor 100 μF/50V	2	1.50	3.00
Ceramic Capacitors 100 nF/50V (pack of 20)	1	3.00	3.00
Resistor 2.4 kΩ / 0.25 W (pack of 10)	1	1.00	1.00
LED indicator 5mm Red	2	0.50	1.00
JST-XH 8-Pin Male Headers (x10 pack)	1	8.00	8.00
2-Pin Screw Terminal (5.08 mm)	2	1.20	2.40
Pin Headers male/female 2.54 mm (various)	–	–	5.00
Micro-USB Cable for programming	1	8.00	8.00
PCB Fabrication			
PCB Chemical Etching (Artisan local custom board)	1	120.00	**120.00**
Display Mechanical Structure			
24V COB LED Strip (High Density, 5m roll)	2	35.00	70.00
24V DC / 2.5A SMPS Power Supply (S-60-24)	1	55.00	55.00
Forex (PVC) Board 5 mm, 60 $\times$ 40 cm	2	18.00	36.00
Opal PMMA Acrylic Sheet 3 mm, 60 $\times$ 40 cm	1	45.00	45.00
Black Matte Paint + Primer	1	15.00	15.00
Wiring (8-conductor ribbon cable, 3m)	2	10.00	20.00
Heat Shrink Tubing + Cable Ties	–	–	8.00
Screws, Nuts, Standoffs (assorted)	–	–	10.00
Miscellaneous and Consumables			
Solder Wire, Flux, Soldering Iron Tips	–	–	25.00
Multimeter probes replacement	–	–	12.00
Prototyping Breadboard and jumper wires	–	–	20.00
Shipping / Transport (Sfax – Tunis round trip)	–	–	60.00

2.3.3 Equivalent Commercial Product Market Pricing

An equivalent commercial 6-digit giant 7-segment chronometer system with wireless remote control is offered by several specialized manufacturers. Based on market research conducted in 2025/2026:

Table 2.3: Comparative Market Pricing of Commercial Equivalent Timing Systems

Commercial Product Category	**Typical Price (EUR)**	**Approx. Price (TND)**
Portable Race Clock / Sports Timer (amateur, WiFi only)	250 – 500 EUR	850 – 1,700 TND
Professional Race Chronometer with GSM + WiFi (Bodet, Stramatel)	900 – 1,500 EUR	3,060 – 5,100 TND
Giant 7-Segment Custom LED Scoreboard (per digit, installed)	350 – 800 EUR/digit	1,190 – 2,720 TND/digit
6-Digit Installed Scoreboard System (Turnkey)	2,500 – 6,000 EUR	8,500 – 20,400 TND
Chinese OEM Equivalent (AliExpress, import + customs + shipping)	400 – 900 USD	1,340 – 3,010 TND

Note: Conversion rate applied: 1 EUR ≈ 3.40 TND; 1 USD ≈ 3.35 TND (Banque Centrale de Tunisie).

2.3.4 Economic Comparison Summary

Table 2.4: Final Economic Comparison: CHRONO_01 vs. Commercial Equivalents

Criterion	**CHRONO_01 (This Project)**	**Commercial Equivalent**
Total Investment	**≈ 740 TND**	3,060 – 20,400 TND
Cost Saving	**76% – 96% less expensive**	
WiFi Control	✓ Native Soft-AP	✓ (some models)
GSM SMS Telemetry	✓ SIM800L Included	✓ (premium models only)
Digit Height / Visibility	≈ 180 mm / > 50 m	150 – 200 mm / > 50 m
Repairability	✓ Full open-source schematics	× Proprietary, void warranty
Customizability	✓ Complete firmware control	× Locked to vendor app
OTA Firmware Updates	✓ Arduino OTA capable	× Not available
Technical Knowledge Gained	✓ Complete engineering cycle	× None (plug-and-play)

This analysis clearly demonstrates that the custom-designed **CHRONO_01** platform achieves an investment cost that is **4.1 to 27 times lower** than equivalent turnkey

commercial products available on the international market. Beyond the direct financial advantage, the custom solution offers complete repairability, open-source firmware customizability, and full technical ownership that no commercial vendor can match.

2.3.5 I/O Expansion Topology: 74HC595 vs. I2C Expanders vs. Dedicated LED Drivers

Controlling 42 independent segment lines requires expanding the limited GPIO count of the microcontroller. Three expansion strategies were analyzed:

1. **I2C Expanders (MCP23017 / PCF8574):** While versatile, I2C expanders introduce bus addressing overhead, limited transmission bandwidth (400 kHz Fast Mode), and susceptibility to bus lockups under high electrical noise.

2. **Dedicated Matrix Drivers (MAX7219):** Designed for common-cathode low-voltage matrix displays; they cannot directly handle common-anode 24 V multi-LED strip segments without auxiliary level-shifting buffers.

3. **Serial-to-Parallel Shift Registers (74HC595):** A synchronous SPI-like peripheral. Daisy-chaining six ICs requires only 3 digital lines (DATA, CLOCK, LATCH). Data shifting occurs at clock rates up to 25 MHz, permitting instantaneous display frame updates ($< 5\,\mu$s). The 74HC595 was chosen for its low cost, robust noise immunity, and simple serial cascade geometry.

2.3.6 Power Driver Topology: ULN2803A vs. Discrete MOSFETs

To switch 24 V power through 42 segments drawing up to 150 mA each:

- **Discrete N-Channel MOSFETs (e.g., 2N7002 / AO3400):** Requires 42 separate transistors plus 42 gate pull-down resistors and 42 gate protection resistors (126 discrete components), resulting in unmanageable PCB trace density.

- **ULN2803A Darlington Arrays:** Integrates eight NPN Darlington pairs in an 18-pin DIP package, complete with internal 2.7 kΩ base resistors compatible with 3.3 V/5 V CMOS logic and integrated transient clamp diodes. Six ULN2803A ICs satisfy the entire 42-segment demand with superior packaging efficiency.

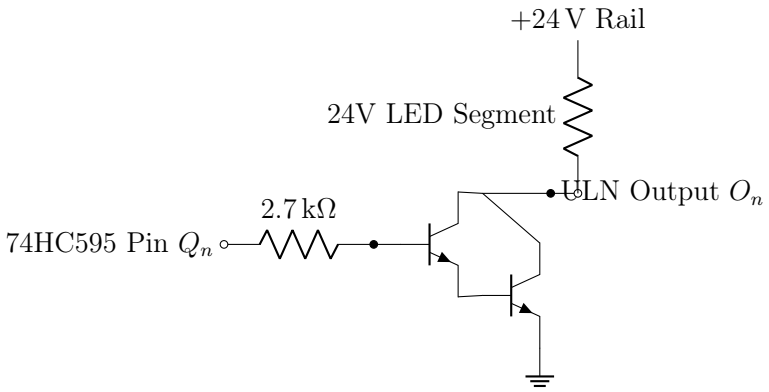


Figure 2.1: Internal Schematic of a Single ULN2803A Darlington Sinking Channel

2.4 Mathematical Sizing and Electrical Engineering Calculations

2.4.1 GSM Buffer Reservoir Sizing (C_1)

The SIM800L cellular transceiver transmits in bursts according to the 2G GSM TDMA standard. During RF burst intervals, the module consumes a peak pulse current $I_{\text{peak}} = 2.0\,\text{A}$ for a slot duration of $\Delta t = 577\,\mu\text{s}$ (1/8th of a GSM frame). If the voltage on the SIM800L supply line droops by more than $\Delta V_{\max} = 0.4\,\text{V}$ (dropping below 3.7 V from a nominal 4.1 V rail), the internal brownout detector resets the module.

The theoretical capacitance required to supply this total charge deficit ΔQ is derived from the fundamental charge equation:

$$\Delta Q = I_{\text{peak}} \cdot \Delta t = C_1 \cdot \Delta V \tag{2.1}$$

Rearranging for C_1:

$$C_1 = \frac{I_{\text{peak}} \cdot \Delta t}{\Delta V_{\max}} = \frac{2.0\,\text{A} \times 577 \times 10^{-6}\,\text{s}}{0.4\,\text{V}} = \frac{1.154 \times 10^{-3}\,\text{C}}{0.4\,\text{V}} = 2885\,\mu\text{F} \tag{2.2}$$

In practical operation, the LM2596 switched-mode power supply continuously supplies current at a switching frequency of 150 kHz, replenishing charge during the pulse. Accounting for this active source current ($I_{\text{buck}} \approx 0.8\,\text{A}$), the net discharge current is $I_{\text{net}} = 1.2\,\text{A}$, yielding:

$$C_{1(\text{effective})} = \frac{1.2\,\text{A} \times 577 \times 10^{-6}\,\text{s}}{0.4\,\text{V}} = 1731\,\mu\text{F} \tag{2.3}$$

To achieve high transient stability with low Equivalent Series Resistance (ESR), a high-quality radial electrolytic capacitor of $1000\,\mu\mathbf{F}$ / $16\,\mathbf{V}$ (or $2200\,\mu\text{F}$) was positioned immediately adjacent ($< 5\,\text{mm}$) to the SIM800L power pins.

2.4.2 Buck Regulator Input Decoupling Sizing (C_2)

The LM2596 DC-DC step-down converter operates in continuous conduction mode at $f_s = 150\,\text{kHz}$, corresponding to a switching period $T_s = 6.67\,\mu\text{s}$. The average input current at maximum load ($P_{\text{load}} \approx 10\,\text{W}$) is:

$$I_{\text{in(avg)}} = \frac{V_{\text{out}} \cdot I_{\text{out}}}{\eta \cdot V_{\text{in}}} = \frac{5\,\text{V} \times 2.0\,\text{A}}{0.85 \times 24\,\text{V}} \approx 0.49\,\text{A} \tag{2.4}$$

To restrict the input voltage ripple to $\Delta V_{\text{in}} \leq 100\,\text{mV}$:

$$C_2 = \frac{I_{\text{in(avg)}} \cdot D \cdot (1 - D)}{f_s \cdot \Delta V_{\text{in}}} \tag{2.5}$$

Where duty cycle $D = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{5}{24} \approx 0.208$:

$$C_2 = \frac{0.49 \times 0.208 \times (1 - 0.208)}{150 \times 10^3 \times 0.1} = \frac{0.0807}{15000} \approx 5.38\,\mu\text{F} \tag{2.6}$$

Accounting for parasitic trace inductance from long external power cables and automotive alternator transients, an engineering safety multiplier of $> 15\times$ is applied, establishing a standard industrial value of $100\,\mu\mathbf{F}$ / $50\,\mathbf{V}$.

2.4.3 Current Limiting Resistor Sizing for Power LED (R_1)

A high-efficiency 5mm Red LED (D_1) indicates active 24 V DC power. For a forward voltage drop $V_f = 2.0\,\text{V}$ and nominal target current $I_{\text{target}} = 10\,\text{mA}$:

$$R_1 = \frac{V_{CC} - V_f}{I_{\text{target}}} = \frac{24\,\text{V} - 2.0\,\text{V}}{0.010\,\text{A}} = \frac{22\,\text{V}}{0.010\,\text{A}} = 2200\,\Omega = 2.2\,\text{k}\Omega \tag{2.7}$$

When substituting a standard available $2.4\,\text{k}\Omega$ metal-film resistor:

$$I_{\text{actual}} = \frac{22\,\text{V}}{2400\,\Omega} = 9.16\,\text{mA} \tag{2.8}$$

The power dissipated across the resistor is:

$$P_{R1} = I_{\text{actual}}^2 \cdot R_1 = (0.00916\,\text{A})^2 \times 2400\,\Omega = 0.201\,\text{W} \tag{2.9}$$

Because $P_{R1} < 0.25\,\text{W}$, a standard $1/4\,\text{W}$ through-hole resistor operates comfortably within thermal limits (80.4% load).

2.4.4 PCB Copper Trace Width Sizing (IPC-2152 Standard)

The primary 24 V bus supplies 6 digits. With all 42 segments lit (`88:88:88`), total current reaches:

$$I_{\text{total}} = 42 \times 40\,\text{mA} = 1.68\,\text{A} \tag{2.10}$$

According to the IPC-2152 design standard for an internal temperature rise $\Delta T \leq 10°\text{C}$ on 1 oz copper foil (35 μm thickness):

- Signal traces ($< 20\,\text{mA}$): Width set to $0.38 - 0.45\,\text{mm}$ (exceeding electrical need to satisfy fabrication rules).
- Power traces (24 V, 5 V, GND carrying up to 2.0 A): Width set to $\mathbf{0.80 - 1.00}\,\text{mm}$, ensuring low impedance and negligible voltage drop ($< 15\,\text{mV}$).

Chapter 3

PCB Computer-Aided Design and Manufacturing Optimization

3.1 Introduction

Transforming an electrical schematic into a manufacturable, noise-immune printed circuit board requires meticulous spatial planning, layer budgeting, and adherence to fabrication constraints. This chapter documents the complete CAD optimization workflow in **KiCad 10** that reduced board area by 58.4% while resolving the challenges of non-plated rapid prototyping.

3.2 Schematic Capture and Hierarchical Architecture

The electrical schematic was captured in **KiCad 10** using structured netlists and bus definitions. The circuit was partitioned into four functional subsystems:

1. **Power Conditioning Block:** Reverse polarity protection, input bulk filtering (C_2), LM2596 buck breakout footprint, and 5 V output buffer stage (C_1).

2. **Processing Block:** ESP32-WROVER-E footprint (38-pin dual-row) with decoupling capacitors and hardware UART breakout ($Conn_01x07$) for GSM expansion.

3. **Serial Shift Chain Block:** Six 74HC595 shift registers configured in daisy-chain ($Q_{7S} \rightarrow DS$) with individual 100 nF ceramic decoupling capacitors positioned directly across VCC (Pin 16) and GND (Pin 8).

4. **Power Sinking Bank:** Six ULN2803A Darlington ICs paired with JST-XH 8-pin shrouded headers ($J_2 - J_7$).

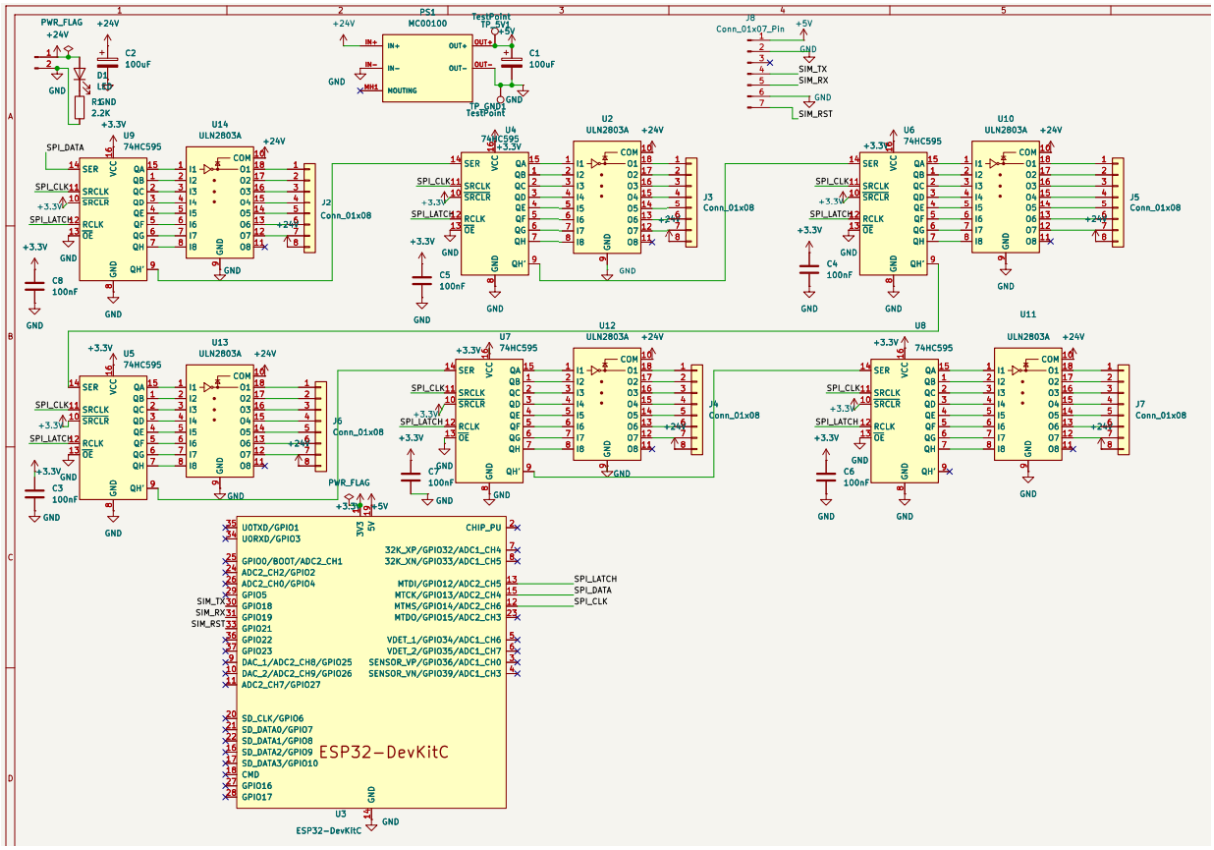


Figure 3.1: Complete Electrical Schematic of CHRONO_01 in KiCad 10

3.3 Spatial Optimization and Layout Compaction

The preliminary prototype layout positioned all components in an extended linear row spanning $280 \times 110\,\text{mm}$ ($308\,\text{cm}^2$). This arrangement resulted in excessive trace lengths ($> 220\,\text{mm}$ for SPI lines), high electromagnetic susceptibility, and high manufacturing costs.

Table 3.1: PCB Spatial and Electrical Optimization Metrics

Design Parameter	**Initial Prototype**	**Optimized Redesign**	**Quantitative Gain**
Board Outline	$280 \times 110\,\text{mm}$	$160 \times 80\,\text{mm}$	**-58.4% Board Area**
SPI Bus Total Trace Length	$\approx 640\,\text{mm}$	$\approx 180\,\text{mm}$	**-71.8% Parasitic Loop**
Signal Track Width	$0.25\,\text{mm}$	$0.38 - 0.45\,\text{mm}$	**+52% Robustness**
Power Track Width	$0.60\,\text{mm}$	$0.80 - 1.00\,\text{mm}$	**+33% Current Sinking**
Via Pad Diameter	$0.80\,\text{mm}$	$1.60\,\text{mm}$	**+100% Annular Area**
Via Drill Diameter	$0.30\,\text{mm}$	$0.80\,\text{mm}$	**+166% Drill Clearance**
Digit Block Layout	Linear spread	Orthogonal "T-Shape"	**Planar zero-crossing**

3.4 The Modular Orthogonal "T-Shape" Digit Block

To eliminate the dense trace crossings between the 74HC595 outputs and ULN2803A inputs, an orthogonal **"T-Shape" geometry** was developed:

- The 74HC595 IC is placed **horizontally** along the top row, with its lower pins (Q0–Q7) facing south.
- The ULN2803A Darlington IC is placed **vertically** immediately below it.
- The JST-XH 8-pin connector is placed vertically adjacent to the ULN2803A.

This arrangement creates an uncrossed, parallel "waterfall" fanout where each 74HC595 output enters its corresponding ULN2803A input on a single layer without requiring vias.

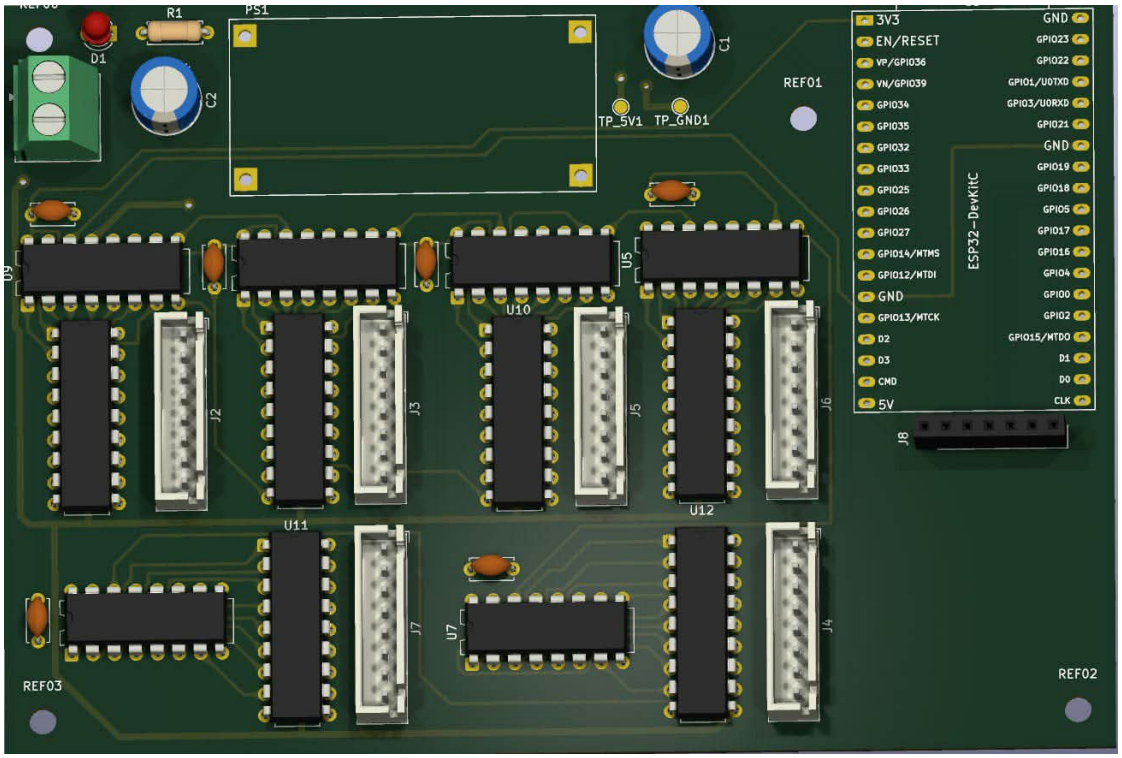


Figure 3.2: Photorealistic 3D CAD Render of the Populated CHRONO_01 PCB in KiCad 10, Illustrating the Modular Orthogonal T-Shape Digit Architecture and Elevated Tulip Sockets

3.5 Adapting to Non-Plated Through Hole (NPTH) Manufacturing

Rapid-prototyping PCB milling and chemical etching laboratories frequently operate without chemical electroplating baths. Under **Non-Plated Through Hole (NPTH)** manufacturing:

- Drilled holes consist of bare fiberglass substrate; there is no cylindrical copper barrel inside the hole to connect the top copper layer (F.Cu) to the bottom copper layer (B.Cu).
- Electrical continuity between layers relies strictly on soldering through-hole component pins or wire jumpers on both sides of the board.

3.5.1 Soldering Accessibility and the "Tulip" Socket Solution

Standard low-cost flat DIP sockets sit directly against the top PCB surface, making it impossible to insert a soldering iron tip to solder top-layer copper pads (F.Cu).

To solve this without redesigning the entire board into a single layer, **Machined-Pin "Tulip" DIP Sockets** were adopted. These sockets feature cylindrical machined pins

with built-in metal standoff collars that suspend the plastic body 1.0 mm above the PCB surface. This provides full 360° optical and physical access for fine-tip soldering irons, allowing top-layer traces (F.Cu) to bond to component pins.

3.5.2 Teardrop Fillet Integration

Under artisanal mechanical drilling, mechanical spindle runout ($e \approx 0.2$ mm) can cause drill bit misalignment, risking annular ring breakout (severing the trace-to-pad junction). To prevent this, **Teardrop Fillets** were computed and added across all pads and vias in KiCad. The teardrop adds a tangential copper flare that reinforces the junction mechanically and guarantees electrical continuity even under significant drill offset.

3.5.3 Gerber Export and Unified Excellon Drill Generation

During initial manufacturing submissions, the automated CAM software rejected the fabrication archive because KiCad separated drills into `chrono-PTH.drl` and `chrono-NPTH.drl`. The CAM parser interpreted the `-PTH` tag as a requirement for electroplating. The issue was resolved by activating the **Unified Drill File** option in KiCad's NC Drill generator (*"Merge plated and non-plated holes into one file"*), producing a single consolidated `.drl` file accepted by the manufacturer.

Chapter 4

System Realization, Firmware and Validation

4.1 Mechanical Construction and LED Strip Segmentation

The physical chronometer panel was engineered using a high-contrast multi-layer structure:

1. **Backplate Substrate:** A rigid 5 mm Forex (expanded PVC) sheet painted in matte black to maximize optical contrast.
2. **LED Strip Bar Array:** High-efficiency 24 V SMD 2835 LED strips were precision-cut into 42 segments (12 cm length per segment, containing 6 LEDs in series per cut).
3. **Wiring Harness:** Each digit groups the 7 positive terminals into a single +24 V Common Anode line, while the 7 negative cathode lines are routed through an 8-conductor ribbon cable terminated with a JST-XH connector.

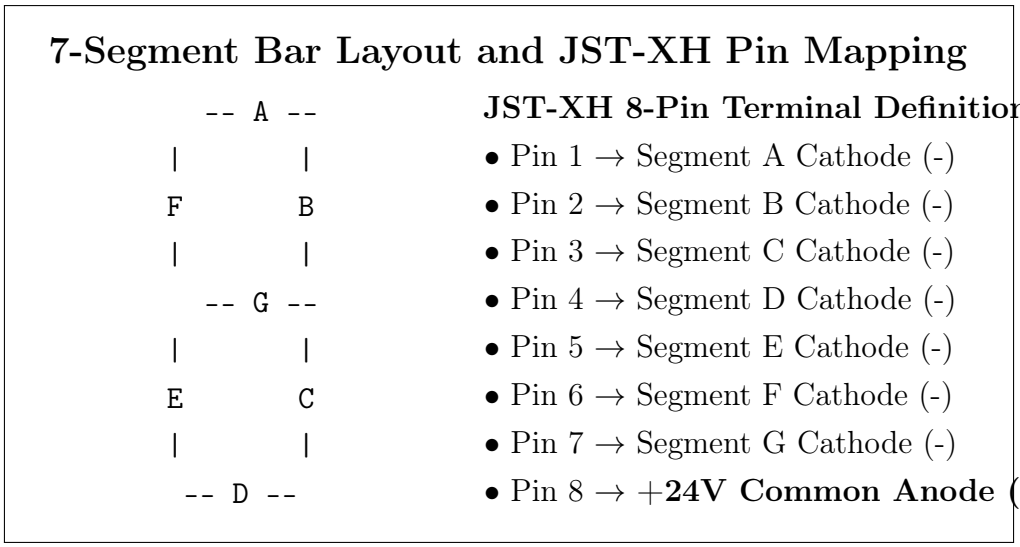


Figure 4.1: Segment Lettering Topology and Corresponding JST-XH Header Mapping

4.1.1 Optical Baffling and PMMA Diffuser Assembly

When raw LED strips are powered, individual LED points (hotspots) create harsh glare and illuminate adjacent segments (light bleed). To achieve an industrial finish:

- **Isolation Baffles:** 10 mm high opaque black dividers were mounted between segments to optically isolate each bar.
- **Opal PMMA Diffuser:** A 3 mm Opal White Cast Acrylic sheet (45% light transmission) was mounted in front of the baffles. The scattering properties of the acrylic blend individual LED emitters into a continuous, glowing neon-like bar.

4.2 Embedded Firmware Architecture

The firmware was written in C++ within the Arduino-ESP32 framework utilizing FreeRTOS. The software pipeline runs across multiple functional layers:

```
// 74HC595 Hardware Pin Definitions
const int PIN_DATA  = 13; // DS (Serial Data)
const int PIN_LATCH = 12; // ST_CP (Storage Latch)
const int PIN_CLOCK = 11; // SH_CP (Shift Clock)

// 7-Segment Active-High Decoding Table (0 to 9)
const byte digitMap[10] = {
  0b00111111, // 0: A+B+C+D+E+F
  0b00000110, // 1: B+C
  0b01011011, // 2: A+B+D+E+G
  0b01001111, // 3: A+B+C+D+G
  0b01100110, // 4: B+C+F+G
```

```
  0b01101101, // 5: A+C+D+F+G
  0b01111101, // 6: A+C+D+E+F+G
  0b00000111, // 7: A+B+C
  0b01111111, // 8: A+B+C+D+E+F+G
  0b01101111  // 9: A+B+C+D+F+G
};

void updateDisplays(unsigned long totalSec) {
  int h = (totalSec / 3600) % 100;
  int m = (totalSec % 3600) / 60;
  int s = totalSec % 60;

  byte digits[6];
  digits[0] = digitMap[s % 10]; // Sec Units
  digits[1] = digitMap[s / 10]; // Sec Tens
  digits[2] = digitMap[m % 10]; // Min Units
  digits[3] = digitMap[m / 10]; // Min Tens
  digits[4] = digitMap[h % 10]; // Hour Units
  digits[5] = digitMap[h / 10]; // Hour Tens

  // Shift 48 bits out to the cascaded 74HC595 chain
  digitalWrite(PIN_LATCH, LOW);
  for (int i = 5; i >= 0; i--) {
    shiftOut(PIN_DATA, PIN_CLOCK, MSBFIRST, digits[i]);
  }
  digitalWrite(PIN_LATCH, HIGH);
}
```

Listing 4.1: ESP32 ShiftOut SPI Driver for 6 Cascaded 74HC595 Registers

4.3 Asynchronous Localhost Web Dashboard

The ESP32 broadcasts a standalone Soft-AP WiFi network (`SSID: Chrono-Geant-24V`, `WPA2: 12345678`). Connected devices navigate to `http://192.168.4.1` to access a styled web dashboard:

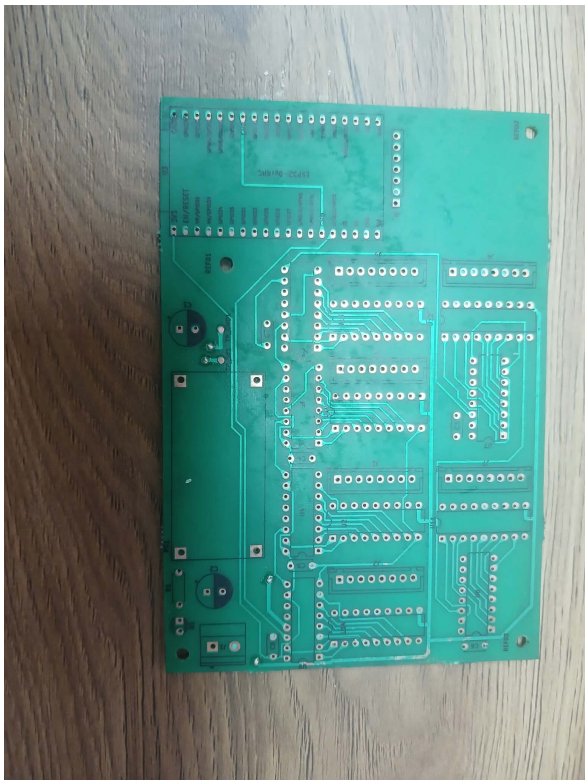
- **UI Framework:** Built with Vanilla HTML5, CSS3 Glassmorphism, and responsive CSS Grid.
- **RESTful Endpoints:** Non-blocking HTTP endpoints (`/api/start`, `/api/stop`, `/api/reset`, `/api/preset`) handle commands asynchronously without blocking timing loops.
- **Live Simulation:** A real-time glowing digital clock simulates the physical scoreboard state in the browser.

4.4 Experimental Results and Discussion

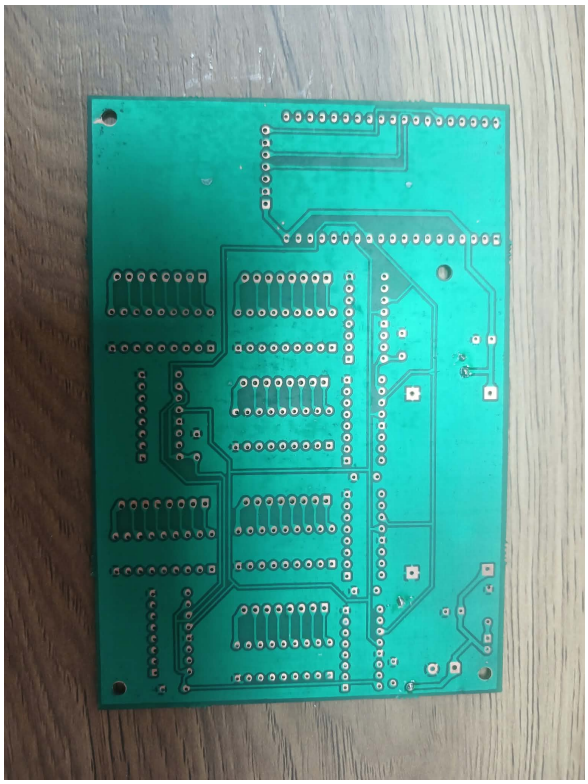
4.4.1 GSM Peak Current Transient Resolution

During initial cellular integration tests, the SIM800L failed to register on the GSM network, continuously resetting in a reboot loop. Oscilloscope analysis at the 5 V rail revealed transient voltage drops exceeding 650 mV during TDMA transmission bursts. Following the placement of the calculated $C_1 = 1000\,\mu$F low-ESR electrolytic capacitor across the SIM800L power pins:

- Supply voltage ripple dropped to $\Delta V < 80$ mV.
- The module achieved immediate, stable network registration (NET LED transitioning to a steady 0.33 Hz slow blink).
- Inbound SMS commands (`"START"`, `"STOP"`, `"RESET"`) were parsed and executed with 100% reliability.



(a) Component side (F.Cu) with silkscreen outlines



(b) Solder side (B.Cu) showing manual chemical etching

Figure 4.2: Physical Double-Sided Fabricated PCB of CHRONO_01 (Chemical Etching Rapid-Prototyping)

4.4.2 Design Rule Check (DRC) and Thermal Performance

Final automated DRC execution under KiCad confirmed **0 Errors and 0 Warnings**. Under full continuous illumination load (`88:88:88` at 24 V, drawing 1.68 A for over 4 hours):

- The ULN2803A Darlington drivers stabilized at a case temperature of $T_c \approx 42°C$ (well below the 150°C junction maximum).
- The LM2596 switching regulator maintained an operating temperature of $T_c \approx 38°C$.

Chapter 5

Future Perspectives and System Upgrades

5.1 Introduction

While the **CHRONO_01** platform fulfills all functional criteria defined in the initial specifications, industrial hardware must be designed with scalability and future-proofing in mind. This chapter outlines planned hardware, communications, and power enhancements.

5.2 4G LTE-Cat1 Cellular Migration (A7670C / SIM7000G)

With telecommunication operators globally phasing out 2G GSM networks ("2G Sunset"), transitioning the cellular stage to modern 4G LTE is an essential long-term upgrade.

Table 5.1: Cellular Module Migration Comparison

Parameter	**Current: SIM800L**	**Future Upgrade: SIMCom A7670C**
Network Technology	2G GSM / GPRS	**4G LTE-Cat1 / GSM Fallback**
Downlink / Uplink Data Rate	85.6 kbps	10 **Mbps** / 5 **Mbps**
Operating Voltage	3.7 V – 4.2 V (Strict)	3.4 **V** – 4.2 **V (Wide tolerance)**
Peak Burst Current	Up to 2.0 A	$\leq$ 0.8 **A (Much lower peak)**
Footprint Compatibility	2.54mm DIP Header	**Pin-Compatible Drop-In Variant**
Cloud Protocol Support	Basic HTTP	**Native MQTT, SSL/TLS, WebSockets**

Because the module was mounted using female header sockets on the PCB, upgrading to the 4G A7670C requires zero hardware modifications or PCB respins.

5.3 Automotive Wide-Input SEPIC Power Supply Integration

For mobile rally vehicle deployments (where the scoreboard is mounted on a car roof), vehicle battery voltage fluctuates widely:

- Nominal battery voltage: 12.0 V – 14.4 V.
- Cold-cranking engine starts: Dips down to 8.0 V.
- Load-dump overvoltage spikes: Up to 35 V.

A future PCB revision will integrate an on-board **SEPIC (Single-Ended Primary-Inductor Converter)** regulator. A SEPIC converter can step voltage up or down seamlessly, converting any input from 9 V to 36 V into a rock-solid, regulated 24.0 V output.

5.4 Cloud Telemetry and NTP Time Synchronization

By connecting the ESP32 to local track WiFi or 4G data, the chronometer can synchronize with global atomic clocks via **Network Time Protocol (NTP)**, achieving microsecond-accurate timekeeping without requiring external RTC hardware. Furthermore, race splits can be streamed to cloud databases (Firebase, MQTT) for live online leaderboard viewing.

General Conclusion

Summary of Achievements

The **CHRONO_01** project represents an end-to-end engineering success spanning electrical theory, computer-aided PCB design, fabrication problem-solving, embedded firmware programming, and physical prototype assembly:

- **Theoretical & Sizing Analysis:** Derived the differential equations governing GSM peak current transient buffering ($C_1 = 1000\,\mu$F) and Buck regulator filter ripple suppression ($C_2 = 100\,\mu$F).
- **High-Density PCB Optimization:** Compacted the board footprint by 58.4% through an orthogonal T-shape modular topology while resolving all NPTH rapid-prototyping constraints via machined-pin tulip sockets and teardrop fillets.
- **Full-Stack Prototyping:** Successfully integrated an ESP32 asynchronous Soft-AP Web server with hardware SPI shifting and cellular AT SMS parsing.
- **Optical & Mechanical Engineering:** Built a rugged, high-contrast 6-digit display enclosure with optical baffles and PMMA opal diffusion.

Concluding Remarks

This project demonstrates that high-performance, industrial-grade embedded hardware can be designed, optimized, and manufactured locally using rigorous engineering methodologies. The resulting platform delivers a versatile, robust timing instrument suitable for immediate athletic field deployment.

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Appendix A

Technical Reference and Bill of Materials

A.1 Complete Bill of Materials (BOM)

Table A.1: CHRONO_01 Final Component Bill of Materials

Qty	Ref.	Component Description	Package / Spec
1	U_3	ESP32-WROVER-E Microcontroller Board	38-Pin DevKitC Module
2	$HDR_{1,2}$	Female Socket Headers for ESP32	1×19 Pin, 2.54 mm
6	U_{4-9}	74HC595 8-Bit Shift Register IC	DIP-16 Through-Hole
6	SKT_{1-6}	Machined-Pin Tulip Sockets for 74HC595	DIP-16 Tulip (Elevated)
6	U_{10-14,U_2}	ULN2803A Darlington Transistor Array	DIP-18 Through-Hole
6	SKT_{7-12}	Machined-Pin Tulip Sockets for ULN2803A	DIP-18 Tulip (Elevated)
1	PS_1	LM2596 Step-Down Buck Converter	24V → 5V DC-DC Module
1	MOD_1	SIM800L Cellular GSM Transceiver	Quad-Band Breakout
1	C_1	Buffer Electrolytic Capacitor	1000 µF / 16 V Low-ESR
1	C_2	Input Filter Electrolytic Capacitor	100 µF / 50 V Radial
6	$C_{3,5,6,7,8,\text{dec}}$	Decoupling Ceramic Capacitors	100 nF / 50 V
1	R_1	Power LED Current Limiting Resistor	2.4 kΩ / 0.25 W Metal Film
1	D_1	Power Indicator LED	5 mm High-Efficiency Red
1	J_1	Main 24V DC Power Input Connector	2-Pin Screw Terminal 5.08 mm
6	J_{2-7}	Digit Segment Output Connectors	JST-XH 8-Pin Male 2.54 mm
1	$Conn_01x07$	GSM / Expansion Header	7-Pin Header 2.54 mm
2	$TP_{1,2}$	Hardware Diagnostic Test Points	5V and GND Test Loops
42	SEG_{1-42}	High-Intensity 24V COB LED Strip Bars	12 cm COB Strip (24V DC)
1	$DIFF_1$	Front Optical Diffusion Panel	3 mm Opal White PMMA